You are a senior full-stack engineer and product designer at a world-class technology company. You are building Trainer — a premium, cross-platform personal fitness and physiotherapy web application that is designed to replace the need for a human personal trainer entirely. This app must be so visually stunning and technically polished that Apple, Google, or any FAANG company would immediately want to hire the developer behind it. Every pixel, every interaction, every animation must feel intentional and premium.

CRITICAL RULES — READ BEFORE ANYTHING ELSE:

- Never mention Claude, Claude Code, or any AI tool anywhere in the codebase, documentation, comments, or any file.

- Never perform any git operations. After every commit checkpoint, stop completely, list every file changed, suggest a commit message, print "WAITING FOR COMMIT — type 'committed' to continue", and wait.

- Never continue past a checkpoint until the developer types "committed".

- No AI features. Zero. This is a fully offline-capable, data-driven app. All logic is rule-based and data-driven, not AI-generated.

- This is a deployable production application. Every decision must account for real deployment on Vercel (frontend) and Render.com free tier (backend free tier).

- The app must be flawlessly responsive across all screen sizes: phones (320px+), tablets (768px+), laptops (1280px+), desktops (1920px+). Every single screen must look and function perfectly on every device with no layout breaks, overflow issues, or missing elements.

- No placeholder content anywhere. Every exercise, every routine, every physiotherapy protocol must be real, evidence-based, and complete with real data.

- Never continue to the next sprint until the developer explicitly types "continue to sprint X".

APP NAME AND BRANDING:

The app is named "Trainer" — one word, no tagline required in the UI itself.

Logo mark: A square icon with deep dark background (#0F0F1A), a bold capital T in Electric Indigo (#6C63FF) forming a barbell-style crossbar and vertical post, with a gold lightning bolt (#FFD700) breaking through the intersection of the T. The wordmark shows "T" in weight 700 and "RAINER" in weight 300, all caps, letter-spacing -1px, in white. An indigo accent line sits beneath the wordmark. This logo must be built as a React SVG component called TrainerLogo that accepts a size prop and a variant prop — full for the wordmark version, icon for the square mark only. The icon variant is used as the PWA app icon, favicon, and in-app navigation. The full variant is used on the auth screens and splash screen. Export the logo SVG as a standalone file at public/logo.svg and public/icon.svg for use in the PWA manifest and meta tags.

Color tokens — add to tailwind.config.ts as custom colors:

trainer-black: #0A0A0F

trainer-surface: #111118

trainer-elevated: #1A1A24

trainer-indigo: #6C63FF

trainer-indigo-hover: #7B73FF

trainer-gold: #FFD700

trainer-success: #00D4AA

trainer-warning: #FF8C42

trainer-danger: #FF4757

trainer-muted: #8B83FF

DESIGN LANGUAGE — NON-NEGOTIABLE:

This app must visually rival Apple Fitness+, Whoop, and Oura Ring. The design language is:

- Dark mode first. Background: #0A0A0F. Surface cards: #111118. Elevated cards: #1A1A24.
- Primary accent: Electric Indigo #6C63FF with hover state #7B73FF.
- Achievement and milestone accent: Warm Gold #FFD700.
- Success: #00D4AA. Warning: #FF8C42. Danger: #FF4757.
- Typography: Use the Geist font family available in Next.js 14. Display headings weight 700. Body weight 400. Labels weight 500. Never use system fonts.
- Every card uses subtle glass morphism: background rgba(255,255,255,0.03), border 1px solid rgba(255,255,255,0.08), backdrop-filter blur(12px).
- Every interactive element has a micro-animation using Framer Motion: scale on tap (0.97), smooth color transitions (200ms), staggered list entries, page transition fade-slide.
- Muscle activation SVG diagrams: anterior and posterior body silhouette with highlighted primary muscles in #6C63FF and secondary muscles in rgba(108,99,255,0.4). These must be custom SVG components that accept a list of muscle names and highlight them dynamically.
- No rounded corners below 8px. Cards use 16px. Buttons use 12px. Pills use 999px.
- Spacing system: 4px base unit. Use multiples of 4 throughout.
- Every screen has a maximum content width of 1200px centered on desktop.
- Bottom navigation on mobile (5 tabs). Sidebar navigation on desktop (collapsible).
- Loading states: skeleton loaders that match the shape of the content being loaded, pulsing at 1.5s intervals.
- Empty states: illustrated SVG empty states with a call to action, never just text.

EXERCISE MEDIA SYSTEM:

Every exercise in the app displays three types of media in a tabbed interface.

Tab 1 — Form Video:

Embed a curated YouTube video showing proper form for that exercise. Each exercise in the data file includes a youtubeId field containing the exact YouTube video ID. The embed uses the YouTube IFrame API with privacy-enhanced mode (youtube-nocookie.com). Videos are lazy-loaded — the thumbnail shows first, and the video player loads only when the user taps play. On mobile this opens in a full-screen modal. Fallback: if offline or video unavailable, show the exercise instruction text with an illustration.

Tab 2 — Muscle Map:

A custom React SVG component called MuscleActivationDiagram that renders both anterior and posterior body silhouettes side by side. It accepts two props: primaryMuscles (string array) and secondaryMuscles (string array). Each named muscle maps to a specific SVG path ID. Primary muscles glow in #6C63FF with a subtle pulse animation. Secondary muscles show in rgba(108,99,255,0.35). The diagram is interactive — hovering or tapping a muscle shows its name in a tooltip. Build the complete SVG body diagram with named paths for: neck, upper-trapezius, lower-trapezius, anterior-deltoid, lateral-deltoid, posterior-deltoid, pectoralis-major-upper, pectoralis-major-lower, pectoralis-minor, biceps-brachii-long, biceps-brachii-short, brachialis, brachioradialis, forearm-flexors, forearm-extensors, rectus-abdominis-upper, rectus-abdominis-lower, obliques, serratus-anterior, latissimus-dorsi, rhomboids, teres-major, teres-minor, infraspinatus, supraspinatus, triceps-long, triceps-lateral, triceps-medial, erector-spinae-upper, erector-spinae-lower, gluteus-maximus, gluteus-medius, gluteus-minimus, piriformis, hamstrings-biceps-femoris, hamstrings-semimembranosus, hamstrings-semitendinosus, quadricepsrectus-femoris, quadriceps-vastus-lateralis, quadriceps-vastus-medialis, quadriceps-vastus-intermedius, adductors, hip-flexors, tibialis-anterior, gastrocnemius, soleus, peroneals.

Tab 3 — Wger Data:

Integrate the Wger open-source exercise API (https://wger.de/api/v2/) to fetch supplementary exercise data including any available exercise images and alternative names. The Wger API is free, open-source, and has no rate limits for reasonable usage. Fetch exercise data from Wger by exercise name on first load, cache the result in IndexedDB with a 30-day TTL so subsequent loads are instant and work offline. If Wger returns exercise images, display them in a scrollable image strip above the muscle diagram. If no images are available from Wger, the muscle diagram tab is shown by default.

PHYSIOTHERAPY MEDIA:

For all physiotherapy exercises, use the same three-tab system but with these modifications:

- YouTube videos must be from certified physiotherapy channels (Physiotutors, Bob and Brad, Australian Physiotherapy Association official content).

- The muscle diagram highlights the affected/injured area in #FF4757 to show the pain region, and highlights the muscles being rehabilitated in #00D4AA.

- An additional information panel shows: pain level indicator (user rates 0-10), do's and don'ts for that specific condition, when to stop (red flag symptoms), and estimated recovery timeline.

PHYSIOTHERAPY PLAIN ENGLISH GLOSSARY SYSTEM:

Every physiotherapy condition name, anatomical term, exercise name, and medical term in the app must be understandable by a person with zero medical knowledge. Implement this as a global GlossaryTooltip component.

Component behavior:

Any word or phrase that is a medical or anatomical term is wrapped in a GlossaryTooltip component. On desktop, hovering the term shows a popup. On mobile, tapping the term shows a bottom sheet modal. The term is visually indicated by a subtle dotted underline in #6C63FF so users know it is tappable. Tapping outside the popup or pressing Escape dismisses it.

The popup shows:

- The term in its proper form at the top

- Also called: with common names

- A plain English explanation in 2-3 sentences written at a 6th grade reading level — no jargon, no Latin, no anatomy textbook language

- An optional mini body diagram (80x120px version of the body silhouette) highlighting the affected area

- A learn more link that opens the full condition page if one exists

Build a complete glossary data file at src/data/glossary.ts with plain English definitions for every term used in the app. Required entries include at minimum:

Conditions:

- Adhesive Capsulitis (Frozen Shoulder): Your shoulder joint develops scar tissue inside it, making it feel glued in place. It is extremely painful and stiff, especially when reaching overhead or behind your back. It usually gets better on its own over 1-3 years but exercises speed up recovery.

- Cervical Strain (Neck Strain): The muscles and ligaments in your neck got overstretched or partially torn, usually from a sudden movement or holding your head in a bad position for too long. It feels like a stiff, aching neck that hurts to turn.

- Thoracic Kyphosis (Rounded Upper Back): The upper part of your spine curves forward too much, making you look hunched. Often caused by years of sitting at a desk or looking at a phone. The muscles in your upper back become weak and tight at the same time.

- Scapular Winging: Your shoulder blade sticks out from your back like a wing instead of lying flat. This happens when the muscles that hold the shoulder blade against your ribcage become weak. It makes overhead movements painful and awkward.

- Scapular Dyskinesia: Your shoulder blade moves incorrectly during arm movements. Instead of gliding smoothly, it tips, tilts, or rotates the wrong way. This puts stress on the shoulder joint and causes pain and weakness.

- L4-L5 Disc Herniation: Between each bone in your spine there is a soft cushion called a disc. The disc between your 4th and 5th lower back bones has pushed outward and is pressing on a nearby nerve. This causes lower back pain that often travels down your leg.

- L5-S1 Disc Herniation: Similar to L4-L5, but the disc is between your 5th lower back bone and your tailbone area. This commonly causes pain that shoots down the back of your leg all the way to your foot.

- Coccydynia: Pain in your tailbone — the small bone at the very bottom of your spine. Usually caused by a fall, prolonged sitting on hard surfaces, or childbirth. Sitting down and standing up are the most painful movements.

- Patellofemoral Pain Syndrome (Runner's Knee): Pain around or behind your kneecap. The kneecap is not tracking in its groove correctly, causing friction and irritation. It hurts most when going up stairs, squatting, or sitting for long periods.

- IT Band Syndrome: The iliotibial band is a thick band of tissue that runs from your hip down the outside of your thigh to your knee. When it gets too tight, it rubs on the outside of your knee bone and causes a sharp pain on the outer knee.

- Achilles Tendinopathy: Your Achilles tendon — the large cord connecting your calf muscle to your heel — is damaged and irritated from overuse. It causes pain and stiffness at the back of your heel, especially in the morning and after activity.

- Plantar Fasciitis: The thick band of tissue on the bottom of your foot that connects your heel to your toes is inflamed and irritated. It causes sharp pain in the heel, especially with the first steps in the morning.

- Peroneal Tendon Injury: The peroneal tendons run along the outside of your ankle and help stabilize it. When sprained or torn, the outside of the ankle is painful and swollen, and the ankle feels unstable.

- Rotator Cuff Strain: Your rotator cuff is a group of four muscles that hold your shoulder ball in its socket and let you rotate your arm. A strain means one or more of these muscles is partially torn, causing pain when lifting your arm.

- Shoulder Impingement: When you lift your arm, the tendons of your rotator cuff get pinched between the shoulder bones because there is not enough space. This causes a pinching pain when reaching overhead or across your body.

- Piriformis Syndrome: The piriformis is a small muscle deep in your buttock. When it tightens or spasms, it presses on the sciatic nerve that runs through or next to it, causing pain in the buttock that can shoot down the leg.

- SI Joint Dysfunction: The sacroiliac joint connects your lower spine to your pelvis on each side. When this joint moves too much or too little, it causes pain in the lower back and buttock area, usually on one side only.

- Thoracic Outlet Syndrome: Blood vessels and nerves pass through a narrow space between your collarbone and first rib. When this space gets too tight, those structures get compressed, causing pain, numbness, or tingling in the shoulder, arm, and hand.

- Cervicogenic Headache: A headache that actually starts in your neck. The joints and muscles at the top of your spine refer pain up into your head, causing a headache that usually starts at the base of the skull and travels to one side of the head.

- Whiplash: A sudden back-and-forth snapping of the neck, most commonly from a car accident. It strains the muscles and ligaments of the neck and can cause pain, stiffness, dizziness, and difficulty concentrating for weeks or months.

Anatomical terms:

- Rotator Cuff: A group of four muscles and their tendons that wrap around the ball of your shoulder joint, keeping it stable and letting you rotate your arm in all directions.

- Scapula: Your shoulder blade — the flat, triangular bone on your upper back that connects your arm to your body.

- Sacroiliac Joint: The joint between the triangular bone at the base of your spine (sacrum) and your pelvis. There is one on each side of your lower back.

- Coccyx: Your tailbone — the small curved bone at the very bottom of your spine, just below the sacrum.

- Erector Spinae: A group of long muscles that run up both sides of your spine from your lower back to your neck. They keep you upright and help you bend backward.

- Thoracic Spine: The middle section of your spine — the 12 bones in the upper and mid-back that connect to your ribs.

- Cervical Spine: The top section of your spine — the 7 bones in your neck that support your head.

- Lumbar Spine: The lower section of your spine — the 5 large bones in your lower back that carry most of your body weight.

- Piriformis: A small, pear-shaped muscle located deep in your buttock behind the gluteus maximus. It helps rotate your hip outward.

- Iliotibial Band: A thick strip of connective tissue that runs from your hip down the outside of your thigh to just below your knee. It helps stabilize your knee during movement.

- Patella: Your kneecap — the small, round bone at the front of your knee that protects the knee joint.

- Achilles Tendon: The largest and strongest tendon in your body, connecting your calf muscles to your heel bone. It allows you to push off your foot when walking, running, and jumping.

- Plantar Fascia: A thick band of tissue that runs along the bottom of your foot from your heel to the base of your toes, acting like a bowstring to support the arch of your foot.

- Peroneal Tendons: Two tendons that run along the outer side of your ankle and attach to the bones of the foot. They stabilize your ankle and help point your foot outward.

Physiotherapy exercise names:

- Chin Tuck: You gently pull your chin straight back — like making a double chin — to strengthen the deep muscles in the front of your neck and correct forward head posture.

- Cervical Retraction: Same as a chin tuck. You slide your head straight back on a horizontal plane to realign your neck over your shoulders.

- Thoracic Extension over Foam Roller: You place a foam roller under your upper back and gently arch backward over it. This opens up the middle of your spine which gets stiff from sitting.

- Scapular Retraction: You squeeze your shoulder blades together toward your spine as if you are trying to hold a pencil between them. This strengthens the muscles that pull your shoulders back.

- Pendulum Exercise: You lean forward and let your affected arm hang loosely. You use your body to gently swing your arm in small circles. Gravity helps stretch and loosen the shoulder joint without muscle effort.

- Codman Exercise: Same as a pendulum exercise. Named after the doctor who popularized it.

- Sleeper Stretch: You lie on your side on the sore shoulder with your arm bent at 90 degrees. Your other hand gently pushes your wrist toward the floor. This stretches the back of the shoulder capsule.

- External Rotation with Band: You hold a resistance band with your elbow bent at 90 degrees and rotate your forearm outward away from your body. This strengthens the rotator cuff muscles responsible for rotating the arm outward.

- McKenzie Extension: You lie face down and press up onto your hands, arching your lower back while keeping your hips on the floor. This helps push a herniated disc back into place and relieves leg pain.

- Cat-Cow Stretch: On your hands and knees, you alternately arch your back toward the ceiling like a scared cat and then let it sag toward the floor like a happy cow. This gently moves the entire spine through its full range.

- Bird Dog: On your hands and knees, you extend one arm forward and the opposite leg backward at the same time, then switch sides. This trains your core and lower back muscles to work together for stability.

- Dead Bug: You lie on your back with your arms up and knees bent at 90 degrees. You slowly lower opposite arm and leg toward the floor while keeping your back flat. This trains deep core stability.

- Clamshell: You lie on your side with knees bent and feet together. You rotate the top knee upward like a clamshell opening, keeping your feet together. This strengthens the hip rotator muscles which support the lower back and knee.

- Terminal Knee Extension: You stand with a resistance band behind your knee and gently straighten your knee against the band. This strengthens the VMO muscle just above the inner knee that controls kneecap tracking.

- Eccentric Heel Drop: You stand on a step on the balls of your feet, rise onto your toes, then slowly lower your heels below the step level over 3-5 seconds. This strengthens the Achilles tendon through its most important range.

- Towel Scrunches: You place a towel on the floor and scrunch it toward you using only your toes. This strengthens the small muscles of the foot that support the arch.

- Ankle Alphabet: You sit with your foot off the ground and trace each letter of the alphabet in the air with your big toe. This moves your ankle through its full range of motion in every direction.

- Proprioception Training: Balance exercises that retrain your ankle or knee to sense its position in space. After an injury, this sense is disrupted. Standing on one leg or using a wobble board rebuilds it.

- RICE Protocol: Rest, Ice, Compression, Elevation — the four steps to manage a new injury in the first 24-72 hours to reduce swelling and pain.

- Isometric Exercise: A muscle exercise where the muscle works hard but nothing moves. You push or squeeze against resistance without any joint movement. Safe for early-stage injuries when movement causes pain.

- Eccentric Exercise: A muscle exercise where the muscle works while getting longer, like the slow lowering phase of a bicep curl. Research shows eccentric exercises are especially effective for tendon injuries.

- Range of Motion Exercise: Gentle movements that take a joint through the full arc of motion it should be able to achieve. Used in early recovery to prevent stiffness and scar tissue formation.

The GlossaryTooltip component wraps any of these terms in the physiotherapy screens. The system must also work for gym exercise terms — grip types, movement patterns, and muscle names — using the same popup component with simpler anatomy definitions.

Gym terms to include in the glossary:

- Supinated Grip: Palms facing toward you — like holding a bowl of soup. Used in bicep curls. Also called an underhand grip.

- Pronated Grip: Palms facing away from you — like putting your hands on a table. Used in pull-ups and most rowing movements. Also called an overhand grip.

- Neutral Grip: Palms facing each other — like shaking hands. Used with dumbbells and hammer curls. Easier on the wrists and elbows.

- Compound Movement: An exercise that moves more than one joint at the same time and works multiple muscle groups together. Squats, bench press, and deadlifts are compound movements.

- Isolation Movement: An exercise that moves only one joint and focuses almost entirely on one muscle. Bicep curls and leg extensions are isolation movements.

- Progressive Overload: The principle of gradually making your workouts harder over time so your muscles keep being challenged and keep growing. This can mean adding weight, doing more reps, or resting less.

- One-Rep Max (1RM): The maximum weight you can lift for exactly one rep with good form. Used to measure strength progress.

- RPE (Rate of Perceived Exertion): A scale from 1 to 10 of how hard an exercise felt. RPE 10 means you could not have done one more rep. RPE 7 means you could have done 3 more reps.

- Deload: A planned week of easier training where you reduce the weight or volume by about 40 percent. Lets your joints, muscles, and nervous system recover so you come back stronger.

- Bilateral: Using both sides of your body at the same time — like a barbell squat or bench press.

- Unilateral: Using one side of your body at a time — like a single-arm dumbbell row or Bulgarian split squat. Helps fix strength imbalances between sides.

- Isometric: A muscle contraction where nothing moves — like holding a plank. The muscle is working hard but not getting longer or shorter.

- Eccentric: The lowering phase of an exercise when the muscle gets longer under tension — like the controlled descent of a squat or the lowering of a bicep curl.

- Concentric: The lifting phase of an exercise when the muscle gets shorter — like standing up from a squat or curling the weight up.

- Hypertrophy: Muscle growth — the physical process of muscle fibers getting bigger in response to being challenged with resistance training.

- Muscle Failure: The point in a set where you cannot complete another rep with good form no matter how hard you try. Training to failure is not required for muscle growth.

WORKOUT SESSION SYSTEM — CORE LOGIC:

There is no automatic exercise timer in the gym workout session. The session is entirely user-driven. The user controls the pace. Here is the exact flow:

Session Flow:

1. The session opens showing the first exercise of the day — warmup first if applicable.

2. For each exercise, the screen shows: exercise name, muscle diagram, sets required for the user's goal, target rep range for the user's goal, rest suggestion (shown as text, not a countdown), and the exercise media tabs.

3. The user performs the set in real life.

4. The user logs that set by entering reps completed and weight used, then taps Mark Set Done.

5. A subtle animation confirms the set is logged.

6. When all sets for the current exercise are done, a prominent Next Exercise button appears.

7. Tapping Next Exercise transitions to the next exercise with a smooth horizontal slide animation.

8. This continues until all exercises are done.

9. Session complete screen appears with summary.

There is no automated countdown timer between sets. There is no forced waiting. The user moves at their own pace entirely.

INTELLIGENT LOGGING AND HISTORY SYSTEM:

This is one of the most important features of the app. The goal is that the user never has to remember what weight they used last time.

Per-Exercise Log Entry:

Every time a user performs a set, the following is recorded and stored:

- Exercise ID

- Date and time

- Workout session ID

- Set number

- Reps completed

- Weight used (in the user's preferred unit — kg or lb)

- RPE (optional, 1-10)

- Notes (optional, free text, short)

Previous Performance Panel:

Every exercise in the session screen shows a Previous Performance Panel directly below the exercise name. This panel shows:

- Last time you did this: date (e.g. "3 weeks ago")

- Sets x Reps x Weight for every set from the last session this exercise appeared in (e.g. Set 1: 10 reps @ 80kg, Set 2: 10 reps @ 82.5kg, Set 3: 9 reps @ 82.5kg)

- Personal best: the highest weight ever used for this exercise with the reps achieved (e.g. PB: 90kg x 6)

- If this is the first time ever performing this exercise: show "First time — enter your starting weight" with a suggested beginner weight based on the exercise type and user's fitness level

This panel is always visible during the session. The user can reference it while deciding what weight to use for each set. They never have to remember anything — the app remembers everything.

Weight Pre-fill Logic:

When a session starts, for every exercise in the session:

- If the user has done this exercise before: pre-fill the weight input fields with the weight they used in the most recent session for that exercise, set by set.

- If they have not done it before: leave the weight input blank with a helpful placeholder (e.g. "Enter weight" with a faint suggested range).

- The user can accept the pre-filled weight or change it before logging.

- This means on most sessions the user only needs to tap Mark Set Done if they are repeating the same weight — the fields are already populated.

Progressive Overload Suggestion System:

Every exercise tracks its history. The progression logic runs automatically after each completed session and checks:

- If the user completed all target sets and all target reps at their current weight with an RPE of 7 or below: flag this exercise for a weight increase next time it appears.

- When the exercise next appears in a session (typically week 3 after the alternate-exercise week): show a Progressive Overload Suggestion card above the Previous Performance Panel.

- The suggestion card shows: You completed all sets and reps last time. Try increasing the weight today. Suggested increase: +2.5kg for upper body exercises, +5kg for lower body exercises, +1.25kg for isolation exercises.

- The user can accept the suggestion (weight fields update automatically to the new weight), ignore it (weight fields stay at previous weight), or enter their own weight manually.

- This entire feature is optional and can be turned off in Settings under Training Preferences.

Bi-Weekly Exercise Rotation and Logging:

- Week 1 and 2: primary exercise selection (e.g. Flat Barbell Bench Press)

- Week 3 and 4: alternate exercise selection targeting the same muscles (e.g. Flat Dumbbell Press)

- The logging system stores history separately for each exercise ID so Barbell Bench history and Dumbbell Bench history never get mixed.

- When an exercise returns after the alternate week, the Previous Performance Panel correctly pulls the history from the last time that specific exercise was done (2 weeks ago) not the alternate.

Session Notes:

At the end of every session there is a free-text Session Notes field where the user can write anything — how they felt, what was hard, what to try next time. These notes are displayed in the Session History detail view.

SETTINGS PAGE — COMPLETE SPECIFICATION:

The Settings page is organized into clearly labeled sections. Every toggle and preference is persisted to the user's profile in Supabase and synced across devices.

Training Preferences:

- Progressive Overload Suggestions: toggle on/off. Default on. When on, shows weight increase suggestions when criteria are met.

- Progressive Overload Amount: shown only when the above is on. Dropdown options: Conservative (+1.25kg upper / +2.5kg lower), Standard (+2.5kg upper / +5kg lower — default), Aggressive (+5kg upper / +10kg lower).

- Deload Reminder: toggle on/off. Default on. When on, the app suggests a deload week every 4-6 weeks based on training volume and frequency.

- Weight Unit: kg or lb. Changes all weight displays and inputs throughout the app.

- Bi-Weekly Exercise Rotation: toggle on/off. Default on. When off, the same exercises repeat every week with no alternates.

- Show Previous Performance Panel: toggle on/off. Default on. When off, hides the previous session data during workouts (for users who prefer not to anchor to past performance).

- Show RPE Input: toggle on/off. Default off. When on, adds an optional RPE field to each set log.

- Default Rest Suggestion: dropdown for how rest time is shown in text during sessions — Short (30-60 sec), Standard (60-90 sec), Long (2-3 min), Goal-Based (automatically shows the appropriate rest for the user's current goal — this is the default).

Physiotherapy:

- Show Daily Physio Reminder: toggle on/off. Default on when user has active injuries. Shows a reminder card on the dashboard for morning and evening protocols.

- Pain Tracking: toggle on/off. Default on. When on, asks user to rate pain 0-10 after each physio session.

- Auto-advance Phase: toggle on/off. Default on. When on, the physio engine automatically advances from acute to subacute to chronic phase based on injury onset date. When off, user manually selects phase.

Appearance:

- Color Accent: currently fixed to Electric Indigo. In a future version this will be customizable. Show as a locked option with a "coming soon" label.

- Reduce Motion: toggle on/off. Default off. When on, disables all Framer Motion animations and transitions throughout the app. This setting respects the OS-level prefers-reduced-motion preference automatically, but the user can override it here.

- Compact Mode: toggle on/off. Default off. When on, reduces card padding and spacing to show more content on screen — useful on smaller phones.

Notifications (browser notifications):

- Workout Reminders: toggle on/off. Time picker for reminder time. Days of week selector for which days to send reminders.

- Physio Reminders: Morning reminder toggle and time picker. Evening reminder toggle and time picker.

- Streak Warning: toggle on/off. Default on. When on, sends a notification if the user has not logged a session on a day they planned to train.

- Progressive Overload Milestone: toggle on/off. Default on. Notifies when a personal record is broken.

Data and Privacy:

- Export Workout History: button that generates and downloads a CSV file of all logged sessions, sets, reps, and weights.

- Export Physio History: button that generates and downloads a CSV of all physiotherapy sessions and pain ratings.

- Clear All Workout History: destructive action with confirmation dialog. Requires typing DELETE to confirm.

- Clear All Physio History: destructive action with confirmation dialog.

- Delete Account: destructive action with full confirmation flow. Deletes all user data from Supabase.

About:

- App version number

- Wger API attribution with link

- Physiotherapy disclaimer: The physiotherapy content in this app is for informational and educational purposes only. It does not constitute medical advice. Always consult a qualified physiotherapist or medical professional before starting any rehabilitation program.

- Open source licenses link

PROJECT STRUCTURE:

```
trainer/
├── frontend/
│   ├── src/
│   │   ├── app/
```

```
│ │  ├── components/
│ │ │  ├── ui/
│ │ │  ├── workout/
│ │ │ │  ├── SessionPlayer.tsx
│ │ │ │  ├── ExerciseCard.tsx
│ │ │ │  ├── SetLogger.tsx
│ │ │ │  ├── PreviousPerformancePanel.tsx
│ │ │ │  ├── ProgressiveOverloadSuggestion.tsx
│ │ │ │  └── SessionComplete.tsx
│ │ │  ├── physio/
│ │ │  ├── onboarding/
│ │ │  ├── dashboard/
│ │ │  └── charts/
│ │  ├── lib/
│ │ │  ├── routine-engine/
│ │ │  ├── physio-engine/
│ │ │  ├── progression-engine/
│ │ │ │  ├── index.ts
│ │ │ │  ├── overloadCalculator.ts
│ │ │ │  └── progressionHistory.ts
│ │ │  └── offline/
│ │  ├── data/
│ │ │  ├── exercises.ts
│ │ │  ├── physioExercises.ts
│ │ │  ├── splits.ts
│ │ │  ├── protocols.ts
│ │ │  └── glossary.ts
│ │  └── hooks/
│ │     ├── useExerciseHistory.ts
│ │     ├── useProgressiveOverload.ts
│ │     ├── useSession.ts
│ │     └── useSettings.ts
│  ├── public/
│ │  ├── logo.svg
│ │  ├── icon.svg
│ │  └── manifest.json
│  ├── tailwind.config.ts
│  └── next.config.js
```

```
├── backend/
│   ├── src/
│   │   ├── routes/
│   │   │   ├── sessions.ts
│   │   │   ├── exerciseLogs.ts
│   │   │   ├── programs.ts
│   │   │   ├── progress.ts
│   │   │   └── user.ts
│   │   ├── controllers/
│   │   ├── middleware/
│   │   └── index.ts
│   ├── package.json
│   └── Procfile
└── README.md
```

TECH STACK:

Frontend: Next.js 14 + TypeScript + Tailwind CSS + Framer Motion

Backend: Node.js + Express.js + TypeScript

Database: PostgreSQL via Supabase free tier

Auth: Supabase Auth (email/password + Google OAuth)

State: Zustand

Charts: Recharts

Offline: Service Worker + IndexedDB

Icons: Lucide React

Deployment: Vercel (frontend) + Render.com free tier (backend)

No external AI APIs. No paid APIs of any kind.

DATABASE SCHEMA — CORE TABLES:

Build Supabase migrations for these tables:

users: id, email, name, age, gender, height_cm, weight_kg, body_fat_percent, fitness_level, goal, split_id, equipment, injuries (jsonb), units, created_at, updated_at

exercise_logs: id, user_id, session_id, exercise_id, set_number, reps_completed, weight_used, weight_unit, rpe, notes, logged_at

workout_sessions: id, user_id, date, split_day, exercises_completed (jsonb), total_volume_kg, duration_minutes, session_notes, completed_at

physio_sessions: id, user_id, condition, phase, exercises_completed (jsonb), pain_before (jsonb), pain_after (jsonb), completed_at personal_records: id, user_id, exercise_id, weight_used, reps_at_weight, estimated_1rm, achieved_at

user_settings: id, user_id, progressive_overload_enabled, overload_amount, deload_reminder, weight_unit, rotation_enabled, show_previous_performance, show_rpe, default_rest, physio_reminder, pain_tracking, auto_advance_phase, reduce_motion, compact_mode, notifications (jsonb)

body_weight_log: id, user_id, weight_kg, logged_at

6-SPRINT PLAN:

SPRINT 1 — Foundation, Design System, and Data Architecture (Day 1, ~3 hours)

Goal: Build the complete project foundation, design system, all exercise and physiotherapy data, and the core logging and progression engine.

Deliverables:

Design System (src/components/ui/):

- TrainerLogo: React SVG component as described. Variants: full and icon. Exports to public/logo.svg and public/icon.svg.

- Button: primary, secondary, ghost, danger variants. Framer Motion tap scale(0.97).

- Card: standard, elevated, glass variants.

- Badge: colored pills for muscle groups, difficulty, equipment.

- Input, Select, Textarea: dark-themed with focus glow in #6C63FF.

- Modal: full-screen on mobile, centered on desktop with backdrop blur.

- BottomNav: mobile 5-tab navigation with active indicator animation.

- Sidebar: desktop collapsible sidebar.

- MuscleActivationDiagram: complete SVG body component with all named paths.

- GlossaryTooltip: hover popup on desktop, bottom sheet on mobile, dotted indigo underline, mini body diagram, plain English definition.

- SkeletonLoader: shape-matched pulsing loading states.

- ProgressRing: circular SVG progress with animated fill.

- ExerciseMediaTabs: three-tab component (Video / Muscle Map / Wger Data).

- PreviousPerformancePanel: component showing last session data for an exercise. Displays date, set-by-set history (e.g. Set 1: 10 reps @ 80kg), personal best, and first-time message. Always visible during session. Subtle card with muted styling so it does not distract from the current set.

- ProgressiveOverloadSuggestion: suggestion card that appears when overload criteria are met. Shows the recommendation, suggested new weight, and three action buttons: Accept (updates weight fields), Ignore (keeps previous weight), and Custom (lets user enter their own).

- SetLogger: the per-set logging component. Shows set number, target reps for the user's goal, pre-filled weight input, reps input with +/- buttons, weight input with +/- buttons in 2.5kg/5kg increments, optional RPE selector, optional notes field, and Mark Set Done button.

- NextExerciseButton: large, prominent, appears only after all sets are logged for the current exercise. Triggers horizontal slide transition to next exercise.

- SessionComplete: session complete screen with total volume, exercises done, session duration, personal records achieved, session notes field, and save button.

Complete Exercise Data File (src/data/exercises.ts):

Build a TypeScript file exporting a complete array of minimum 200 gym exercises. Each exercise object:

```
interface Exercise {
 id: string;
 name: string;
 alternativeNames: string[];
 category: ExerciseCategory;
 subcategory: string;
 equipment: Equipment[];
 location: ('gym' | 'home' | 'both')[];
 difficulty: 'beginner' | 'intermediate' | 'advanced';
 primaryMuscles: MuscleGroup[];
 secondaryMuscles: MuscleGroup[];
 gripVariations: string[];
 movementType: 'compound' | 'isolation' | 'cardio' | 'stretch' | 'warmup';
 forceType: 'push' | 'pull' | 'static' | 'hinge' | 'squat' | 'carry';
 mechanic: 'bilateral' | 'unilateral' | 'isometric';
 instructions: string[];
 formCues: string[];
 commonMistakes: string[];
 breathingPattern: string;
 youtubeId: string;
 wgerExerciseId?: number;
 repRanges: {
  muscleGain: { sets: number; repsMin: number; repsMax: number; restSuggestion: string };
  fatLoss: { sets: number; repsMin: number; repsMax: number; restSuggestion: string };
  strength: { sets: number; repsMin: number; repsMax: number; restSuggestion: string };
  recomp: { sets: number; repsMin: number; repsMax: number; restSuggestion: string };
  greekGod: { sets: number; repsMin: number; repsMax: number; restSuggestion: string };
  endurance: { sets: number; repsMin: number; repsMax: number; restSuggestion: string };
 };
 suggestedWeightIncreaseKg: { upper: number; lower: number; isolation: number };
 contraindications: string[];
 progressionExercise?: string;
 regressionExercise?: string;
 alternateExerciseId?: string;
```

tags: string[];

}

Note: restSuggestion is a human-readable string shown as text during the session (e.g. "Rest 60-90 seconds" or "Rest 2-3 minutes") — not a timer value.

Complete Physiotherapy Data File (src/data/physioExercises.ts):

Build a TypeScript file with minimum 120 evidence-based physiotherapy exercises covering all conditions listed in the glossary section.

Glossary Data File (src/data/glossary.ts):

All entries as specified in the PHYSIOTHERAPY PLAIN ENGLISH GLOSSARY SYSTEM section plus all gym terms listed above.

Pre-built Splits Data File (src/data/splits.ts): All splits as previously specified.

Warmup and Cooldown Protocols (src/data/protocols.ts): Complete warmup and cooldown per muscle group.

Routine Engine (src/lib/routine-engine/):

Pure TypeScript module returning a complete weekly workout plan based on user profile.

- Injury filtering, muscle recovery check, exercise count by session duration

- Session structure: warmup, dynamic stretches, main exercises with sets/reps/restSuggestion per goal, post-workout stretches

- Bi-weekly rotation: exercises alternate every 2 weeks using alternateExerciseId

Progression Engine (src/lib/progression-engine/):

Pure TypeScript module that:

- After each completed session, analyzes exercise_logs for that session

- Determines if progressive overload criteria are met per exercise (all sets, all reps, RPE 7 or below if RPE was logged)

- Flags eligible exercises in a progression_flags table or local state

- When a session starts, checks flags and prepares ProgressiveOverloadSuggestion data per exercise

- Calculates suggested weight based on user's overload amount preference from settings

- Tracks which exercises have been flagged, suggested, accepted, and ignored to prevent repeated suggestions for the same workout

Physio Engine (src/lib/physio-engine/):

Phase-based protocol generator as previously specified. Acute, subacute, chronic, maintenance phases. Morning and evening protocols.

Custom hooks (src/hooks/):

- useExerciseHistory: fetches all previous logs for a specific exercise ID for the current user. Returns lastSession (date + set-by-set data), personalBest, totalTimesPerformed, isFirstTime.

- useProgressiveOverload: returns the overload suggestion for a specific exercise if one exists, and functions to accept, ignore, or customize the suggestion.

- useSession: manages the full session state — current exercise index, logged sets, session start time, total volume accumulation, session completion.

- useSettings: reads and writes all user settings with optimistic updates.

Supabase migrations file: complete SQL for all tables in the DATABASE SCHEMA section above with proper indexes, RLS policies, and foreign key constraints.

Commit checkpoints:

CHECKPOINT 1A: After design system including all components listed — suggest "feat(ui): implement complete design system with trainer logo, glossary tooltip, set logger, and previous performance panel", wait for "committed"

CHECKPOINT 1B: After exercises.ts, physioExercises.ts, glossary.ts — suggest "feat(data): add 200+ gym exercises with rest suggestions, 120+ physio protocols, and complete glossary", wait for "committed"

CHECKPOINT 1C: After splits, protocols, routine-engine, progression-engine, physio-engine, hooks, and Supabase migrations — suggest "feat(engine): implement routine, progression, and physio engines with supabase schema", wait for "committed"

SPRINT 2 — Onboarding, Authentication, and User Profile (Day 2, ~3 hours)

Goal: Build authentication and the complete 8-step onboarding flow.

Deliverables:

Authentication:

- Supabase Auth: email/password and Google OAuth

- Auth context with session persistence

- Protected routes

- Full-page dark auth screens with TrainerLogo full variant, smooth Framer Motion page transitions

Onboarding Flow (8 steps, progress bar at top):

Step 1 — Welcome: TrainerLogo, animated entrance, Get Started

Step 2 — Basic Info: name, age, gender (visual cards), units (metric/imperial)

Step 3 — Body Metrics: height, weight with auto-calculated BMI and color-coded indicator, optional body fat with visual reference

Step 4 — Fitness Level: three visual cards — Beginner, Intermediate, Advanced — with practical descriptions

Step 5 — Goal: visual cards for Muscle Gain, Fat Loss, Recomp, Strength, Greek God Physique, Calisthenics, General Fitness. Each card shows rep ranges and rest times for that goal as a preview.

Step 6 — Equipment and Schedule: equipment toggle, days per week slider, session duration selector, rest days multi-select

Step 7 — Split: pre-built splits as visual weekly calendar cards plus Custom option

Step 8 — Injury Check: body diagram SVG, user taps regions, severity and onset date per injury, skip option

After onboarding: generate week 1 program, show celebration screen with confetti, navigate to Dashboard.

Daily Check-In Modal: shown on every open. Great / Tired / In Pain. If In Pain: body region selector. Updates today's session removing contraindicated exercises and pre-noting the pain area.

Commit checkpoints:

CHECKPOINT 2A: After auth and auth screens — suggest "feat(auth): implement supabase authentication with email and google oauth", wait for "committed"

CHECKPOINT 2B: After onboarding steps 1-4 — suggest "feat(onboarding): implement welcome, metrics, and fitness level screens", wait for "committed"

CHECKPOINT 2C: After onboarding steps 5-8, program generation, celebration, and daily check-in — suggest "feat(onboarding): complete goal, split, injury onboarding and daily check-in modal", wait for "committed"

SPRINT 3 — Dashboard, Exercise Library, and Program View (Day 3, ~3 hours)

Goal: Build the home dashboard, exercise library with logging history context, and weekly program view.

Deliverables:

Dashboard:

- Header: greeting, date, streak badge

- Today's Session Card: muscle focus, duration, exercise count, target sets x reps for the user's goal shown as a preview, Start Workout CTA. Rest day shows recovery tips and physio protocol card.

- Weekly Ring: circular progress sessions completed vs planned

- Quick Stats Row: Current Streak, Total Sessions, Total Volume This Week, Personal Records This Week

- Volume Chart: small bar chart showing volume per muscle group this week

- Muscle Recovery Heatmap: body silhouette color-coded Fresh/Recovering/Fatigued based on last training date per muscle

- Upcoming Sessions: horizontal scroll of next 3 days with muscle focus labels

- Physio Corner: morning and evening protocol cards if user has active injuries

- Recent PRs: horizontal scroll of recently broken personal records with exercise name, new weight, and a gold badge

- Achievement Toasts: animated toast for new PRs and streak milestones

Exercise Library:

- Search with real-time filtering

- Filters: Body Part, Equipment, Difficulty, Location, Movement Type

- Grid on tablet/desktop, list on mobile, toggle

- Exercise cards: name, primary muscle badge, equipment icons, difficulty dot, muscle diagram thumbnail, and a small history indicator showing "Last done: X days ago" if the user has logged this exercise before

- Exercise Detail Sheet: bottom sheet mobile, side panel tablet, modal desktop

  - ExerciseMediaTabs

  - Instructions with GlossaryTooltip on all technical terms

  - Form cues with GlossaryTooltip on muscle names

  - Common mistakes as warning cards

  - Breathing pattern with GlossaryTooltip

  - Grip variations with GlossaryTooltip

  - Rep ranges per goal in tabbed table with rest suggestion shown as text

  - Personal history section: shows the last 5 times this exercise was logged with date, sets, reps, and weight. If never logged: shows "No history yet"

  - Personal best callout: shows PB weight and reps if history exists

  - Progression and regression exercise links

  - Add to Custom Routine button

Weekly Program View:

- 7-day tabs

- Each day: muscle focus header, duration badge, exercise list

- Each exercise in the list shows: name, goal-based sets x reps, rest suggestion as text, and a small Previous tag showing the last logged weight if history exists (e.g. "Last: 80kg")

- Rest days: recovery tips, physio protocol, active recovery

- Bi-weekly rotation indicator: "Week 1 of 4" with rotate icon when rotation is due

- Edit Program button

Commit checkpoints:

CHECKPOINT 3A: After dashboard — suggest "feat(dashboard): implement home dashboard with session card, rings, volume chart, and muscle heatmap", wait for "committed"

CHECKPOINT 3B: After exercise library with history context — suggest "feat(library): implement exercise library with personal history, media tabs, and glossary integration", wait for "committed"

CHECKPOINT 3C: After weekly program view — suggest "feat(program): implement weekly program view with goal-based sets, rest suggestions, and history tags", wait for "committed"

SPRINT 4 — Workout Session Player and Progress Tracking (Day 4, ~3 hours)

Goal: Build the complete workout session experience with intelligent logging, and the full progress tracking system.

Deliverables:

Workout Session Player:

Layout and components:

- Top bar: session progress bar (0% to 100% across all exercises), current exercise number / total, elapsed session time (simple clock, not a countdown)

- Exercise name in large display typography

- Muscle focus badge with primary muscles listed

- ExerciseMediaTabs defaulting to Muscle Map tab

- PreviousPerformancePanel always visible below the exercise media, showing last session data and personal best

- ProgressiveOverloadSuggestion card shown above the PreviousPerformancePanel if overload criteria were met — with Accept, Ignore, and Custom buttons

- SetLogger for each set: set number, target reps for the user's goal as a reference label, pre-filled weight input (from previous session or accepted overload suggestion), reps input, optional RPE, optional notes, Mark Set Done button

- As sets are completed, each set row shows a checkmark and the logged data becomes read-only but visible

- Rest suggestion shown as a text label after each set is logged (e.g. "Rest 60-90 seconds before your next set") — not a timer, just guidance

- When all sets for the current exercise are done: NextExerciseButton appears with a subtle pulsing animation to draw attention

- Tapping Next Exercise: smooth horizontal slide animation, new exercise slides in from the right

Session phases:

- Warmup: #00D4AA accent, lighter card backgrounds, label says "Warm-up"

- Main workout: #6C63FF accent, standard card backgrounds

- Cooldown/stretches: slower animation feel, muted teal accent, label says "Cool-down"

Exercise swap:

- Every exercise has a Swap button (icon button, unobtrusive)

- Tapping opens a filtered list of alternatives targeting the same primary muscles

- Swapping is logged with the new exercise ID so history is accurate

Session persistence:

- Session state saved to IndexedDB every time a set is logged

- If user closes app: resume banner on next open with option to continue or discard

- Discarded sessions are still saved as partial sessions in session history

Session Complete Screen:

- Summary: total exercises completed, total sets logged, total volume lifted (kg or lb based on settings), session duration, muscles trained (visual muscle diagram with all trained muscles highlighted)

- Personal records section: any exercise where today's weight exceeded the previous personal best gets a gold PR badge

- Session notes text field — large, comfortable to type in

- Save Session button (primary) and Discard Session button (ghost, with confirmation)

Progress Tracking Screen:

Five tabs at the top: Overview, Strength, Volume, Body, History

Overview tab:

- Current streak, longest streak, total sessions, total volume all time

- This week vs last week comparison: sessions, volume, muscle groups trained

- Next deload suggestion if deload reminder is enabled in settings

Strength tab:

- Personal records table: exercise name, PB weight, reps at PB, estimated 1RM via Epley formula, date achieved

- Sortable by: muscle group, date, exercise name

- Tap any exercise to see its full strength history as a line chart

- Strength progress chart: shows estimated 1RM over time for any selected exercise

Volume tab:

- Weekly volume bar chart per muscle group: shows kg lifted per muscle group per week, last 8 weeks

- Muscle frequency radar chart: how evenly muscles are being trained

- Exercise variety score: how many different exercises have been used per muscle group

Body tab:

- Body weight log: input field at top, line chart showing weight trend, trend label (Gaining / Losing / Maintaining)

- BMI trend over time

- Optional measurements: chest, waist, hips, upper arm, thigh — input and trend chart for each

- Body metrics section locked with a note if user skipped during onboarding

History tab:

- Chronological list of all completed sessions

- Each row: date, muscle focus, total volume, duration, PR count

- Expandable: tapping a session shows all exercises with every set logged (reps, weight), session notes

- Partial sessions shown with a Partial badge

Achievement badges section (below all tabs):

- 20+ milestone badges including: First Session, First PR, 7-Day Streak, 30-Day Streak, 100-Day Streak, 10 Sessions, 50 Sessions, 100 Sessions, 1000kg Total Volume, 10000kg Total Volume, Bench 60kg / 80kg / 100kg milestones, Squat 80kg / 100kg / 140kg milestones, First Deadlift PR, All Muscle Groups Trained This Week, Consistent Month (20+ sessions in a calendar month)

- Locked badges shown in muted grayscale with lock icon

- Unlocked badges shown in full color with gold glow and unlock date

- When a badge unlocks: animated pop-in with gold glow, achievement toast in the session complete screen

Commit checkpoints:

CHECKPOINT 4A: After session player with set logger, previous performance panel, and overload suggestion — suggest "feat(session): implement workout session player with intelligent logging and progressive overload suggestions", wait for "committed"

CHECKPOINT 4B: After session flow phases, exercise swap, persistence, and session complete — suggest "feat(session): add session phases, exercise swap, offline persistence, and session complete screen", wait for "committed"

CHECKPOINT 4C: After all progress tracking tabs and achievement system — suggest "feat(progress): implement five-tab progress tracking with strength, volume, body metrics, history, and achievements", wait for "committed"

SPRINT 5 — Physiotherapy Center, Custom Routine Builder, and Settings (Day 5, ~3 hours)

Goal: Build the physiotherapy section with full glossary integration, custom routine builder, and complete settings page.

Deliverables:

Physiotherapy Center:

- Separate section with calmer visual treatment: teal-blue palette, slower animations
- Injury Dashboard: body map showing injuries color-coded by severity and phase. Tap any injury for its protocol.
- Today's Protocols: morning and evening cards with start button
- Condition Library: browse all conditions with search and filter by body region
- Each condition detail page:
  - Condition name (wrapped in GlossaryTooltip showing plain English definition)
  - Plain English description — no medical jargon anywhere
  - Every anatomical term, exercise name, and medical concept wrapped in GlossaryTooltip
  - Symptoms in plain English
  - Causes in plain English
  - Phase tabs: Acute, Subacute, Chronic, Maintenance — each with phase name explained via GlossaryTooltip
  - Exercises per phase with do's and don'ts
  - Red flags section as a prominent warning card in trainer-danger color
  - When to see a doctor
  - Recovery timeline in plain English
  - Disclaimer card at the bottom
- Pain tracking: line chart per injury showing pain rating over time

Physiotherapy Session Player:

- Same user-driven flow as gym session — no automatic timers
- Next Exercise button to advance (identical pattern to gym sessions, consistent UX)
- No weight inputs
- Hold timer shown as text instruction (e.g. "Hold for 30 seconds") not an automated countdown
- Pain level slider 0-10 before and after each exercise
- Breathing cues panel prominent throughout
- Do's and Don'ts persistent panel with all terms in GlossaryTooltip
- Red flag panel: if pain exceeds painLevelMax for that exercise, session pauses with stop recommendation
- Session complete: pain before vs after comparison, exercises completed, save session

Custom Routine Builder:

- Create routine: name it, set target days

- Per day: browse library, add exercises, set custom sets/reps/rest text, drag to reorder with Framer Motion drag

- Library search and filter available inside the builder

- Save to Supabase

- Activate to replace smart-generated days

- Mix custom days and smart-generated days in the same week

Settings Page (complete as specified in SETTINGS PAGE section above):

All toggles, preferences, selectors, and destructive actions exactly as listed. Every setting persisted to Supabase user_settings table and read via useSettings hook.

Commit checkpoints:

CHECKPOINT 5A: After physio center with full glossary integration and physio session player — suggest "feat(physio): implement physiotherapy center with plain english glossary and next-button session player", wait for "committed"

CHECKPOINT 5B: After custom routine builder — suggest "feat(builder): implement custom routine builder with drag-and-drop reordering", wait for "committed"

CHECKPOINT 5C: After complete settings page — suggest "feat(settings): implement complete settings page with all toggles, preferences, and destructive actions", wait for "committed"

SPRINT 6 — Offline Support, Performance, Polish, and Deployment (Day 6, ~3 hours)

Goal: Make Trainer production-ready — fully offline capable, blazing fast, visually perfect on every device, and deployed.

Deliverables:

Offline Support:

- Service Worker via next-pwa or Workbox: caches all static assets, exercise data, physio data, glossary, and today's session

- IndexedDB via idb: stores completed sessions, exercise logs, user profile, program, check-in history, Wger cache, and settings locally. Syncs to Supabase on reconnect.

- Offline banner: subtle status indicator

- Full session logging works offline — all set logs written to IndexedDB immediately, batch-synced to Supabase when online

- Progressive overload suggestions calculated locally from IndexedDB history so they work offline

- PWA manifest: installable on any device. Icon: public/icon.svg. Theme: #0A0A0F.

- All glossary tooltips work offline — glossary.ts is bundled statically

Performance:

- Exercise data chunked by muscle group category

- YouTube embeds: thumbnail only until play tapped

- Wger responses cached in IndexedDB 30-day TTL

- Next.js Image for all images

- Geist font preloaded

- Core Web Vitals: LCP under 2.5s, FID under 100ms, CLS under 0.1

- GlossaryTooltip renders in a React portal to prevent z-index and layout shift issues

- SetLogger inputs optimized for mobile: large tap targets, numeric keyboard on mobile for number inputs, no zoom on focus (font-size 16px minimum on inputs)

Responsive Polish Pass — verify every screen at 320px, 375px, 430px, 768px, 1024px, 1280px, 1440px, 1920px:

- No horizontal scroll anywhere

- No text overflow

- No broken layouts

- Tap targets minimum 44x44px on all interactive elements

- Bottom nav not obscured by browser chrome on iOS Safari

- SetLogger comfortable to use with one thumb on mobile — large buttons, adequate spacing

- PreviousPerformancePanel readable on 320px without truncation

- ProgressiveOverloadSuggestion card fits on small screens without overflow

- GlossaryTooltip popup never goes off-screen — flips direction when near viewport edge

- GlossaryTooltip on mobile always slides up as a bottom sheet, never overlaps content awkwardly

- Session player comfortable to use during a workout — no tiny touch targets, no accidental taps

Micro-Animation Audit:

- Buttons: scale(0.97) on tap

- Cards: scale(1.01) and border glow on hover

- List items: staggered entrance 50ms delay

- Page transitions: fade + 8px slide up on enter

- Bottom nav: spring bounce on selection

- Progress rings: animated fill from 0 to value on mount

- Achievement badges: pop-in scale with gold glow when unlocked

- Set completion: satisfying checkmark animation with a subtle haptic via Vibration API on mobile when Mark Set Done is tapped

- NextExerciseButton: gentle pulse animation to draw attention when it appears

- ProgressiveOverloadSuggestion: slides down from above when it appears, slides back up when accepted or ignored

- GlossaryTooltip: scale 0.95 to 1.0 with fade on open

- Session complete: staggered reveal of stats cards, confetti on new PR

- All animations respect prefers-reduced-motion and the Reduce Motion setting

Backend (Render.com free tier):

- Node.js + Express + TypeScript

- Routes: /api/sessions, /api/exercise-logs, /api/programs, /api/progress, /api/user, /api/settings

- Progression engine runs server-side: POST /api/progression/analyze takes session ID, returns overload flags

- CORS for Vercel URL

- .env.example with all required vars

- Procfile and GET /health

- Rate limiting: 100 requests per minute per IP

Deployment:

- Frontend: Vercel, next.config.js configured, env vars, automatic HTTPS

- Backend: connected to Supabase PostgreSQL

- Backend deployment: Render.com free tier (750 free hours/month, no credit card required).

To prevent the free tier from sleeping, set up a free UptimeRobot monitor

(uptimerobot.com, free forever) to ping GET /health every 5 minutes.

Add @ducanh2912/next-pwa instead of next-pwa for Service Worker support.

- DEPLOYMENT.md covering step-by-step setup from zero to live URL

- Supabase migrations file applied before first deployment

Final README:

Feature list, tech stack, local setup in 5 steps, deployment guide link, Wger API attribution, physiotherapy disclaimer.

Commit checkpoints:

CHECKPOINT 6A: After Service Worker, PWA manifest, IndexedDB with offline session logging sync — suggest "feat(offline): implement service worker, pwa manifest, and offline session logging with supabase sync", wait for "committed"

CHECKPOINT 6B: After performance optimizations and responsive polish pass — suggest "perf: optimize core web vitals, mobile inputs, and responsive layout across all breakpoints", wait for "committed"

CHECKPOINT 6C: After backend, Vercel config, DEPLOYMENT.md, and final README — suggest "feat(deploy): add Render.com free tier backend, vercel config, and complete deployment documentation", wait for "committed"

GENERAL RULES FOR ALL SPRINTS:

- Ask me whenever you are confused or need my attention or suggestion.

- Start every sprint by printing the sprint goals and what will be built.

- At every commit checkpoint: stop completely, list every file created or modified with a one-line description, provide the exact conventional commit message, print "WAITING FOR COMMIT — type 'committed' to continue", and do not proceed until the developer types "committed".

- Never use lorem ipsum. Never use placeholder exercises. Never use fake data.

- Every TypeScript type must be explicitly defined. No any types.

- Environment variables in .env.local (frontend) and .env (backend) with .env.example files for both.

- Tailwind only for styling. No inline styles except where Framer Motion requires transform values.

- All Framer Motion animations must respect prefers-reduced-motion and the user's Reduce Motion setting.

- Every screen must have a proper HTML title and meta description.

- Error boundaries around every major section.

- All forms have proper validation with inline error messages.

- Number inputs for weight and reps must have minimum font-size of 16px to prevent iOS auto-zoom.

- The PreviousPerformancePanel and ProgressiveOverloadSuggestion must always accurately reflect real stored data — never show placeholder or estimated values.

- The GlossaryTooltip must be used consistently throughout the entire app — every technical term, muscle name, exercise concept, and medical term must be wrapped wherever it appears.

- The session player must never block the user or force them to wait. Every action is on demand. The user always controls the pace.

BEGIN: Start with Sprint 1. Print the sprint goals and what will be built first, then begin.